



Jio Business Messaging (JBM) Platform

User and API Integration Guide

v2.3

Contributors:

Name	Role	Company	Email
Senthil Pandian	Product Engineering	Jio Platforms Limited	Senthil.Pandian@ril.com
Shiva Prasad C	Architect	Jio Platforms Limited	Shivaprasad.C@ril.com
Shilpa Popat	Product Management	Jio Platforms Limited	Shilpa1.Popat@ril.com
Shashank Vashist	Product Management	Jio Platforms Limited	Shashank.Vashist@ril.com
Harshit Amar	Product Management	Jio Platforms Limited	Harshit.Amar@ril.com
Anargha Biswas	Technical Lead	Jio Platforms Limited	Anargha.biswas@ril.com

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1. About this document

1.1. Introduction

This document provides an overview of the Jio Business Messaging (hereafter referred to as JBM) Platform and its core feature set. It serves as a comprehensive guide for using JBM APIs, enabling enterprises and service providers to represent their services through Assistants and engage end-users in rich, interactive conversations. Business partners can leverage this platform to build a variety of use cases, including promotional campaigns, notifications, transaction confirmations, and more.

The API documentation includes detailed information on authentication, endpoint structures, request and response formats and error handling, to help developers integrate and utilize the platform effectively.

1.2. Scope

This document is intended to guide business partners to build commercially viable use cases by integrating APIs exposed by JBM to send messages and events to users, upload and manage files, receive messages and events from users etc.

Following are out of scope:

- How business partners build the conversation and assistants
- User experience of end users

1.3. Target Audience

The primary audience for this API documentation includes developers and engineers who will integrate the APIs into their systems, as well as solution architects evaluating technical fit. Product managers and business analysts may also refer to it to understand capabilities and define use cases. Additionally, QA engineers and external partners may use the documentation for testing and integration support.

1.4. Abbreviations

Abbreviations	Description
API	Application Programming Interface
JBM	Jio Business Messaging
RCS	Rich Communication Suite

1.5. URI scheme

For Prod environment, the following URLs should be used.

- For Token Generation:
<https://tgs.businessmessaging.jio.com>

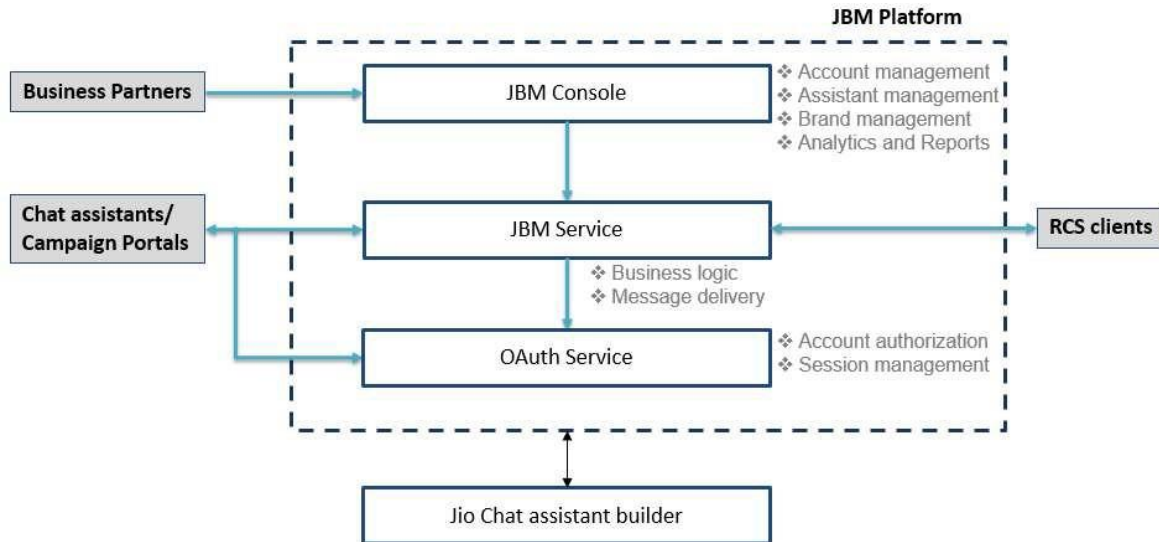
JBM platform uses OAuth 2.0 tokens to securely authorize and authenticate access to its APIs. These tokens act as temporary digital keys that allow users or systems to make API requests without repeatedly sharing sensitive credentials like the Assistant ID and Secret Key.

For security purposes, Expiry time of Oauth token is 3600 secs (1 hour). It needs to be regenerated every 3600 secs (1 hour).

- For Messaging APIs:
<https://api.businessmessaging.jio.com>

2. Architecture

Diagram below explains the high-level architecture of JBM and how business partners or assistant developers could integrate and engage with target users.



3. Message types

Following are the category of messages that are currently served by JBM

Type	Description
Basic	Plain text messages with 160 characters limit and could be visualized to be equivalent to SMS
Rich	Messages with multimedia content, GSMA defined rich cards, carousels, chip lists, quick replies, suggestions and actions or plain text messages over 160 characters
Conversation	When basic or rich messages are responded by end users within a stipulated time, they form a conversation

4. Types of Assistants

There are 3 categories of assistants as would be served by JBM.

Type	Description
Promotional	Assistant that sends messages to end users that are promotional in Nature
Transactional	Assistant that sends transaction confirmation as linked with business backend

OTP	Assistant that sends One-Time-Password (OTP) to confirm authenticity of the user
-----	--

5. IPs to be whitelisted

If at all the clients want to whitelist the IPs from JBM , the Below IPs needs to be whitelisted since the responses and events will be sent towards their webhooks via the below IPs.

34.93.37.102
34.93.217.169
34.93.156.190
34.93.108.116
34.93.238.245
34.93.212.214
34.93.147.9
34.100.142.131
34.47.150.59
35.200.253.4
35.200.244.122
34.100.198.76
35.200.199.7
34.47.192.111
34.47.150.52
34.93.14.3
34.93.189.124
34.100.200.191
34.93.59.70
35.244.7.214
34.47.250.10
35.200.184.114
35.200.209.126
34.100.168.92
34.93.38.186
34.47.200.233
34.47.245.98
35.244.2.62
35.200.155.141
35.244.41.200
35.244.34.186
35.200.255.234
34.47.252.222
34.100.171.236
34.47.143.95
34.100.186.65

34.93.41.87
34.100.142.227
34.93.37.102

6. API Descriptions

6.1. OAuth Token Generation – Get Token

This section defines the API for generating a time-based token from the OAuth server using an Assistant ID and Secret Key.

GET /v1/oauth/token

Parameters

Type	Name	Description	Default	Remarks
Query	grant_type	Method to obtain access token	client_credentials	Constant Value
Query	client_id	Assistant ID	No default value	Every Assistant generates an unique ID upon creation, which is visible in the console.
Query	client_secret	Secret Key	No default value	Users can Secret Key upon assistant creation, which is visible in the console.
Query	scope	Scope of token	Read	Constant Value

Response

HTTP Code	Description	Schema
200	Success	{ "access_token": "MTQYOWE1ODGTNTMWZS0ZZ MEXLW FKN2YTMZFMWWE5ZGM2OTE3", "expires_in": 3600, "scope": "read", "token_type": "Bearer" }
400	Bad Request: The server cannot or will not process the request due to an apparent client error (e.g., malformed request syntax, too large size, invalid	

	request message framing, or deceptive request routing).	
409	Duplicate request	
500	Internal Server Error prevents the server from processing the request.	

6.2. A2P Messaging

This section defines the APIs for sending messages to mobile users from an Assistant.

6.2.1 Send Message

POST

/v1/messaging/users/:userPhoneNumber/assistantMessages/async?messageId={{uid}}&assistantId={{assistantId}}

Description

This API is used to send a message asynchronously from the Assistant to a particular user. In response JBM will send back either SEND_MESSAGE_SUCCESS or SEND_MESSAGE_FAILURE.

Kindly note that we have now removed the restriction of messageId being in uuid format. However, there is a limit of number of characters for messageId. The maximum number of characters for messageId is set to 150 characters.

JBM now supports sending of messageTrafficType as part of the json body. Please note that it is an optional parameter and it can have one of the following five parameters. (AUTHENTICATION, TRANSACTION, PROMOTION, SERVICEREQUEST, ACKNOWLEDGEMENT)

Parameters

Type	Name	Description	Schema	Mandatory/Optional
Path	userPhoneNumber	Recipient's phone number	E.164 format Eg: +914658761230	Mandatory
Query	messageId	Identifier representing the Message.	Max length is 150 characters	Mandatory
Query	assistantId	Identifier of the Assistant used to send the Message		Mandatory

Request Body Parameters

Type	Name	Description	Schema
Object	content	Content of the message	MessageContent [Covered Below Basis Different Message Types]

Basis the different message types, Schemas are covered below:

6.2.1.1 Send Message – Plain Text

In order to send a message with plaintext message to a user, the following schema should be used:

```
{
  "content": {
    "plainText": "This is a plain text message"
  }
}
```

Fields	Description	Parameters
<code>plainText</code>	String Value for the message body.	

6.2.1.2 Send Message – Standalone Rich Card

A standalone rich card is a single rich card, with the following parameters

Fields	Description	Parameters
<code>cardOrientation</code>	Orientation of the card.	- HORIZONTAL - VERTICAL
<code>thumbnailImageAlignment</code>	Direction of image (left/right) in case <code>cardOrientation</code> is HORIZONTAL Not applicable if <code>cardOrientation</code> is VERTICAL	- LEFT - RIGHT
<code>content</code>	Forms the body of the message	Covered Below

Parameters of the content field.

Fields	Description	Mandatory/Optional	Parameters
<code>cardTitle</code>	(string value) (Optional) Title of the card. Maximum 200 characters.	Optional	
<code>cardDescription</code>	(string value) (Optional) Description of the card. Maximum 2000 characters	Optional	
<code>cardMedia</code>	Image, GIF, video , PDF to include in the card.	Optional	- FILEID - FILEURL - THUMBNAIL URL

			MEDIAHEIGHT
mediaHeight	The height of the media within a rich card with a vertical layout. For a standalone card with horizontal layout, height is not customizable	If image media included in card, then mandatory	- SHORT - MEDIUM - TALL
fileUrl	Publicly reachable URL of the file. The content-type field must be present and accurate in the HTTP response from the URL. Recommended maximum file size of 100 MB	If media included in card, one of fileUrl or fileId is mandatory	
fileId	Refer to section 6.2.6 below.		
thumbnailUrl	Publicly reachable URL of the thumbnail. Maximum size of 100 kB.	Optional for video files only	
suggestions	A suggested reply or a suggested action included within a rich card or within a suggestion chip list.	Optional	- REPLY - ACTION
reply	Users can tap a suggested reply to send the text reply back to the Assistant.		
action	Users can tap a suggested action to initiate the corresponding native action on the device.		
plainText	Text that is shown in the label of the Suggested Reply or the Suggested Action	Mandatory [Up to 25 characters]	
postBack	The base64-encoded payload that the Assistant receives in a user event when the user taps the suggested reply or suggested action.	Optional	

Supported media formats in card media

Supported image types:

- JPEG/JPG
- GIF
- PNG

Supported video types:

- H.263
- M4V
- MP4
- MPEG
- MPEG-4
- WebM

Supported file types :

- PDF

Send Message – Horizontal media card

```
{
  "content": {
    "richCardDetails": {
      "standalone": {
        "cardOrientation": "HORIZONTAL",
        "content": {
          "cardTitle": "This is card title",
          "cardDescription": "This is card description",
          "cardMedia": {
            "mediaHeight": "TALL",
            "contentInfo": {
              "fileUrl": "http://www.google.com/logos/doodles/2015/googles-new-logo-5078286822539264.3-hp2x.gif"
            }
          },
          "suggestions": [
            {
              "reply": {
                "plainText": "Suggestion #1",
                "postBack": {
                  "data": "suggestion_1"
                }
              }
            },
            {
              "reply": {
                "plainText": "Suggestion #2",
                "postBack": {
                  "data": "suggestion_2"
                }
              }
            }
          ]
        }
      }
    }
  }
}
```

6.2.1.4.1 Send Message – OpenURL

Plain-Text

```
{
  "content": {
    "plainText": "Visit this URL to find more about our services",
    "suggestions": [
      {
        "action": {
          "plainText": "Browse Now",
          "postBack": {
            "data": "SA1L1C1"
          },
          "openUrl": {
            "url": "https://jio.com"
          }
        }
      }
    ]
  }
}
```

Rich-Card

```
{
  "content": {
    "richCardDetails": {
      "standalone": {
        "cardOrientation": "VERTICAL",
        "content": {
          "cardTitle": "The Street Is About to Meet Its Icon",
          "cardDescription": "A bold new silhouette is emerging. Stay alert—this arrival
will change everything.",
          "cardMedia": {
            "mediaHeight": "MEDIUM",
            "contentInfo": {
              "fileUrl":
"https://jfxv.akamaized.net/PublicStorage/User/6603fe7322e6faffe5a48175/1748512713657.jpeg"
            }
          },
          "suggestions": [
            {
              "action": {
                "plainText": "Know More",
                "postBack": {
                  "data": "SA1L1C1"
                },
                "openUrl": {
                  "url": "https://www.kia.com/in/home.html"
                }
              }
            }
          ]
        }
      }
    }
  }
}
```

OpenUrl

Name	Description	Schema	Required
url	URL string. Must be a valid URI as defined in RFC 3986. Maximum 2048 characters.	string	Y

6.2.1.4.2 Send Message – OpenURL with Webview Plain-Text

```
{
  "content": {
    "plainText": "Visit this URL to find more about Jiosphere",
    "suggestions": [
      {
        "action": {
          "plainText": "Browse Now",
          "postBack": {
            "data": "SA1L1C1"
          },
          "openUrl": {
            "url": "https://medium.com/hprog99/mastering-generics-in-go-a-comprehensive-guide-4d05ec4b12b",
            "application": "WEBVIEW",
            "webviewViewMode": "TALL",
            "description": "its a link"
          }
        }
      }
    ]
  }
}
```

Rich Card

```
{
  "content": {
    "richCardDetails": {
      "standalone": {
        "cardOrientation": "VERTICAL",
        "content": {
          "cardTitle": "The Street Is About to Meet Its Icon",
          "cardDescription": "A bold new silhouette is emerging. Stay alert—this
arrival will change everything.",
          "cardMedia": {
            "mediaHeight": "MEDIUM",
            "contentInfo": {
              "fileUrl":
"https://jfxv.akamaized.net/PublicStorage/User/6603fe7322e6faffe5a48175/1748512713657.jpeg"
            }
          },
          "suggestions": [
            {
              "action": {
                "plainText": "Full Webview",
                "postBack": {
                  "data": "SA1L1C1"
                },
                "openUrl": {
                  "url": "https://amazon.com",
                  "application": "WEBVIEW",
                  "webviewViewMode": "FULL",
                  "description": "its a link"
                }
              }
            }
          ]
        }
      }
    }
  }
}
```

OpenUrl with Webview

Name	Description	Schema	Required
url	URL string. Must be a valid URI as defined in RFC 3986. Maximum 2048 characters.	string	Y
Application	Type of the URL open application. Options: BROWSER	String	Y

	• WEBVIEW		
webviewViewMode	Type of the webview's view mode. Options: • FULL • HALF • TALL	string	Y
description	Accessibility Description for Webview		

6.2.1.2.3 Send Message – Dialer Action

Plain-Text

```
{
  "content": {
    "plainText": "Call this Number to Know More about Jio Assistants",
    "suggestions": [
      {
        "action": {
          "plainText": "Dial Now",
          "postBack": {
            "data": "SA1L1C1"
          },
          "dialerAction": {
            "phoneNumber": "+916367992981"
          }
        }
      }
    ]
  }
}
```

Rich Card

```
{
  "content": {
    "richCardDetails": {
      "standalone": {
        "cardOrientation": "VERTICAL",
        "content": {
          "cardTitle": "The Street Is About to Meet Its Icon",
          "cardDescription": "A bold new silhouette is emerging. Stay alert-this
arrival will change everything.",
          "cardMedia": {
            "mediaHeight": "MEDIUM",
            "contentInfo": {
              "fileUrl":
"https://jfxv.akamaized.net/PublicStorage/User/6603fe7322e6faffe5a48175/1748512713657.jpeg"
            }
          },
          "suggestions": [
            {
              "action": {
                "plainText": "Know More",
                "postBack": {
                  "data": "SA1L1C1"
                },
                "dialerAction": {
                  "phoneNumber": "+916367992981"
                }
              }
            }
          ]
        }
      }
    }
  }
}
```

Dialer Action

Name	Description	Schema	Required
phoneNumber	Phone number	string	Y

6.2.1.2.4 Send Message – Show Location (Lat/Long)

Plain-Text

```
{
  "content": {
    "plainText": "Be Sure to visit this Location",
    "suggestions": [
      {
        "action": {
          "plainText": "Visit Now",
          "postBack": {
            "data": "SA1L1C1"
          },
          "showLocation": {
            "coordinates": {
              "latitude": 21.5937,
              "longitude": 78.9629
            },
            "label": "Label - Show Location"
          }
        }
      }
    ]
  }
}
```

Rich-Card

```
{
  "content": {
    "richCardDetails": {
      "standalone": {
        "cardOrientation": "VERTICAL",
        "content": {
          "cardTitle": "The Street Is About to Meet Its Icon",
          "cardDescription": "A bold new silhouette is emerging. Stay alert—this arrival will
change everything.",
          "cardMedia": {
            "mediaHeight": "MEDIUM",
            "contentInfo": {
              "fileUrl": "https://jfxv.akamaized.net/PublicStorage/User/6603fe7322e6faffe5a
48175/1748512713657.jpeg"
            }
          },
          "suggestions": [
            {
              "action": {
                "plainText": "Know More",
                "postBack": {
                  "data": "SA1L1C1"
                }
              },
              "showLocation": {
                "coordinAtes": {
                  "latitude": 21.5937,
                  "longitude": 78.9629
                },
                "label": "Label - Show Location"
              }
            }
          ]
        }
      }
    }
  }
}
```

Show Location (Lat/Long)

Name	Description	Schema	Required
Latitude	The latitude in degrees. It must be in the range [-90.0, +90.0].	number	Y
Longitude	The longitude in degrees. It must be in the range [-180.0, +180.0].	Number	Y
Label	The label of the pin dropped at latLong.	String	N

6.2.1.2.5 Send Message – Show Location (By Query)

Plain-Text

```
{
  "content": {
    "plainText": "Be Sure to visit this Location",
    "suggestions": [
      {
        "action": {
          "plainText": "Visit Now",
          "postBack": {
            "data": "SA1L1C1"
          },
          "showLocation": {
            "query": "hospitals"
          }
        }
      }
    ]
  }
}
```

Rich-Card

```
{
  "content": {
    "richCardDetails": {
      "standalone": {
        "cardOrientation": "VERTICAL",
        "content": {
          "cardTitle": "The Street Is About to Meet Its Icon",
          "cardDescription": "A bold new silhouette is emerging. Stay alert—this arrival
will change everything.",
          "cardMedia": {
            "mediaHeight": "MEDIUM",
            "contentInfo": {
              "fileUrl":
"https://jfxv.akamaized.net/PublicStorage/User/6603fe7322e6faffe5a48175/1748512713657.jpeg"
            }
          },
          "suggestions": [
            {
              "action": {
                "plainText": "Know More",
                "postBack": {
                  "data": "SA1L1C1"
                },
                "showLocation": {
                  "query": "hospitals"
                }
              }
            }
          ]
        }
      }
    }
  }
}
```

Show Location (Query)

Name	Description	Schema	Required
Query	Location search centered around the user's current location. If the query is sufficiently specific, then Assistants can use it to select any location in the world.	string	Y

6.2.1.2.6 Send Message – Share Location

Plain-Text

```
{
  "content": {
    "plainText": "Click Below button to share your location",
    "suggestions": [
      {
        "action": {
          "plainText": "Share your location",
          "postBack": {
            "data": "SA1L1C1"
          },
          "shareLocation": true
        }
      }
    ]
  }
}
```

Rich-Card

```
{
  "content": {
    "richCardDetails": {
      "standalone": {
        "cardOrientation": "VERTICAL",
        "content": {
          "cardTitle": "The Street Is About to Meet Its Icon",
          "cardDescription": "A bold new silhouette is emerging. Stay alert—this arrival will change everything.",
          "cardMedia": {
            "mediaHeight": "MEDIUM",
            "contentInfo": {
              "fileUrl": "https://jfxv.akamaized.net/PublicStorage/User/6603fe7322e6faffe5a48175/1748512713657.jpeg"
            }
          },
          "suggestions": [
            {
              "action": {
                "plainText": "Share Your Location",
                "postBack": {
                  "data": "SA1L1C1"
                },
                "showLocation": true
              }
            }
          ]
        }
      }
    }
  }
}
```

This type has no parameters.

Opens the RCS app's location chooser so the user can pick a location to send back to the Assistant.

6.2.1.2.7 Send Message – Create Calendar Event

Plain-Text

```
{
  "content": {
    "plainText": "Mark Your Calendar for Assistant Seminar",
    "suggestions": [
      {
        "action": {
          "plainText": "Mark Your Calendar",
          "postBack": {
            "data": "SA111C1"
          },
          "createCalendarEvent": {
            "startTime": "2023-06-26T15:01:23Z",
            "endTime": "2023-06-26T18:01:23Z",
            "title": "Bot Development Seminar",
            "description": "Session 1 of 4"
          }
        }
      }
    ]
  }
}
```

Rich-Card

```

{
  "content":{
    "richCardDetails":{
      "standalone":{
        "cardOrientation":"VERTICAL",
        "content":{
          "cardTitle":"The Street Is About to Meet Its Icon",
          "cardDescription":"A bold new silhouette is emerging. Stay alert—this arrival will
change everything.",
          "cardMedia":{
            "mediaHeight":"MEDIUM",
            "contentInfo":{
              "fileUrl":"https://jfxv.akamaized.net/PublicStorage/User/6603fe7322e6faffe5a48
175/1748512713657.jpeg"
            }
          },
          "suggestions":[
            {
              "action":{
                "plainText":"Mark Your Calendar",
                "postBack":{
                  "data":"SA1L1C1"
                },
                "createCalendarEvent":{
                  "startTime":"2023-06-26T15:01:23Z",
                  "endTime":"2023-06-26T18:01:23Z",
                  "title":"Bot Development Seminar",
                  "description":"Session 1 of 4"
                }
              }
            }
          ]
        }
      }
    }
  }
}

```

Create Calendar Event

Name	Description	Schema	Required
startTime	Event start time. A timestamp in RFC3339 UTC "Zulu" format, with nanosecond resolution and up to nine fractional digits. Examples: "2023-06-26T15:01:23Z" and "2023-06-26T15:01:23.045123456Z"	string	Y
endTime	Event end time. A timestamp in RFC3339 UTC "Zulu" format, with nanosecond resolution and up to nine fractional digits. Examples: "2023-06-26T15:01:23Z" and "2023-06-26T15:01:23.045123456Z".	string	Y
title	Event title.	string	Y
description	Event description.	string	Y

Responses for Send Message API

HTTP Code	Description	Schema
201	Created	{ "success": true, "messageId": {mgid} }
400	Bad Request: The server cannot or will not process the request due to an apparent client error (e.g., malformed request syntax, too large size, invalid request message framing, or deceptive request routing).	
401	Unauthorized: The request requires user authentication or the token needs to be regenerated.	
500	Internal Server Error prevents the server from processing the request.	

Below is an example of message success and message failure events.

Success response type:

```
{
  "userPhoneNumber": "913989xxxx",
  "botId": "bot123",
  "entityType": "STATUS_EVENT",
  "entity": {
    "eventType": "SEND_MESSAGE_SUCCESS",
    "messageId": "msgId1234",
    "sendTime": "2024-09-19",
    "senderPhoneNumber": "913989xxxx",
    "eventId": "eventid1234"
  }
}
```

Failure response type:

```
{
  "userPhoneNumber": "913989xxxx",
  "botId": "bot123",
  "entityType": "STATUS_EVENT",
  "entity": {
    "eventType": "SEND_MESSAGE_FAILURE",
    "error": {
      "code": "internal",
      "errCode": 21,
      "message": "Internal Server Error"
    },
    "messageId": "msgId1234",
    "sendTime": "2024-09-19",
    "senderPhoneNumber": "913989",
    "eventId": "eventid1234"
  }
}
```

The following table depicts mapping of errCodes and their corresponding descriptions.

Message	errCode	Description/ Resolution
Bad Request	2	Request is not proper. Please provide valid json body
Unauthorized	3	Unauthorized.
Forbidden	4	Assistant is not launched yet. Please launch the assistant before retrying again.
User Not found	5	Targeted Phone number may be RCS Disabled or Assistant has not yet been launched/suspended or Targeted user may have registered with Airtel DND
Too many Requests	6	Please check if the max number of messages that can be sent for that particular number from a particular assistant is exhausted for the month.
Internal Server Error	7	This can happen due to various reasons related to Server connectivity Errors. Please retry after sometime
Bot Rate Limit Exceeded	9	Account Level TPS limit Exceeded. Please retry with TPS equals to or lesser than allocated TPS.
Internal Server Error	11	Request Timeout happened. Please try after sometime.
Internal Server Error	12	Bad Gateway Error. Please try after sometime.
Internal Server Error	13	Service Temporarily Unavailable. Please try after sometime.
Internal Server Error	14	Similar to Bad gateway Error. Please try after sometime.
Invalid value for message, one of fields for expiry is already set	15	Please try with suitable expiry fields
Invalid value for message field expireTime	16	Please try with suitable expireTime
Invalid value for message field ttl, the ttl duration must a digit string ending with 's'	17	Please try with suitable expireTime

Internal Server Error	20	GetCapabilities Failed. Please try after sometime.
Internal Server Error	21	SendMessage Request Failed. Please validate the message and try again.
Operation not allowed. Account inactive	23	Account is inactive or balance of this prepaid account is exhausted. Activate/Recharge the account and try again.
Operation not allowed. User has opted out!	24	User has opted out.
Internal Server Error	25	Assistant Not Ready Yet
Too many requests! Account configured TPS exceeded/	28	Account Level TPS Exceeded. Please retry with lower rate.
Too many requests! Default Account TPS exceeded		
Internal Server Error	30	Assistant Initialization Failed. Please re-create assistant and try again.

6.2.2 Revoke Message

DELETE /v1/messaging/users/:userPhoneNumber/assistantMessages/:messageID

Description

This API is used to revoke a message already sent from the Assistant to a particular user.

Response

HTTP Code	Description	Schema
201	Success	{ "success": true, "messageId": uuid }
400	Bad Request: The server cannot or will not process the request due to an apparent client error (e.g., malformed request syntax, too large size, invalid request message framing, or deceptive request routing).	
401	Unauthorized: The request requires user authentication.	
409	Duplicate request	
500	Internal Server Error prevents the server from processing the request.	
404	Not found: Phone number not found in any RCS server	

6.2.3 Events

POST /v1/messaging/users/:userPhoneNumber/assistantEvents

Description

This API is used to send an event from the assistant to a particular user. Partners are encouraged to use these events to provide an engaging user experience to users.

Parameters

Type	Name	Description	Schema	Default
Path	userPhoneNumber	recipient's phone number	E.164 format	No Default Value
Query	eventId	UUID representing the assistant event	UUID	No Default Value

Request Body Parameters

Type	Name	Description	Required
string	eventType	enum (TYPING, MESSAGE_READ)	Y
string	messageId	Applicable only for MESSAGE_READ	N

Response

HTTP Code	Description	Schema
201	Created	{ "success": true, "eventId": uuid }
400	Bad Request: The server cannot or will not process the request due to an apparent client error (e.g., malformed request syntax, too large size, invalid request message framing, or deceptive request routing).	
401	Unauthorized: The request requires user authentication.	
500	Internal Server Error prevents the server from processing the request.	
404	Not found: Phone number not found in any RCS server	

6.2.4 Get user capabilities

GET /v1/messaging/users/:userPhoneNumber/capabilities

Description

Get JBM related capabilities of user. This API provides user's RCS capability and associated information.

Parameters

Type	Name	Description	Schema	Default
Path	userPhoneNumber	recipient's phone number	E.164 format	No Default Value
Query	requestId	UUID representing the assistant event	UUID	No Default Value

Response

HTTP Code	Description	Schema
200	Success	Capabilities
400	Bad Request: The server cannot or will not process the request due to an apparent client error (e.g., malformed request syntax, too large size, invalid request message framing, or deceptive request routing).	
401	Unauthorized: The request requires user authentication.	
403	Forbidden: The request could not be entertained because assistant from which this request is triggered is not launched.	
404	Not Found – User can't be reached through JBM	
500	Internal Server Error prevents the server from processing the request.	

Sample Response:

```
{
  "features": [
    "RICHCARD_STANDALONE_SUPPORTED",
    "CREATE_CALENDAR_EVENT_ACTION",
    "DIAL_ACTION",
    "OPEN_URL_ACTION",
    "SHARE_LOCATION_ACTION",
    "VIEW_LOCATION_ACTION",
    "RICHCARD_CAROUSEL_SUPPORTED",
    "ACTION_OPEN_URL_IN_WEBVIEW"
  ]
}
```

ENUMS	Description
FEATURE_UNSPECIFIED	Not Specified
RICHCARD_STANDALONE_SUPPORTED	Stand alone rich cards are supported
CREATE_CALENDAR_EVENT_ACTION	Suggested action to create a calendar event
DIAL_ACTION	Suggested action to dial a number
OPEN_URL_ACTION	Suggested action to open a URL in browser
SHARE_LOCATION_ACTION	Suggested action to share a location
VIEW_LOCATION_ACTION	Suggested action to view a location in a map app
RICHCARD_CAROUSEL_SUPPORTED	Carousel of rich Cards
ACTION_OPEN_URL_IN_WEBVIEW	Suggested action to open a URL in webview

6.2.5 Batch get users capabilities.

POST /v1/messaging/usersBatchGet

Description

Checks the RCS capability information of list of users. As per Google guidelines, the maximum rate at which batchCapabilities api can be called is 600 requests per minute.

Batch capabilities api to be called for MSISDNs within the range of 500 to 10000 per request.

The returned payload contains a list of RCS-enabled phone numbers reachable by the JBM platform for the specified users. Only phone numbers that are RCS-enabled for a carrier the agent is launched on will be returned.

Request Body Parameters

Type	Name	Description	Required
String[]	phoneNumbers	String array of phone Numbers in E.164 format	Y

Response Status Codes

HTTP Code	Description	Schema
200	Success	
400	Bad Request: The server cannot or will not process the request due to an apparent client error (e.g., malformed request syntax, too large size, invalid request message framing, or deceptive request routing).	
401	Unauthorized: The request requires user authentication.	
403	Forbidden: The request could not be entertained because assistant from which this request is triggered is not launched.	
500	Internal Server Error prevents the server from processing the request.	

Response Body

Name	Description	Schema
reachableUsers	String array containing user's phone numbers that are reachable from the Assistant	string[]

totalRandomSampleUserCount	Count of phone numbers randomly selected from reachableUsers . Typically, this value is ~75% of the total requested phone numbers. This value is 0 if the count of requested phone numbers is less than 500.	integer
reachableRandomSampleUserCount	Count of phone numbers from the randomly selected list that are RCS-reachable across all carriers, regardless of the Assistant's launch status. The ratio between this value and totalRandomSampleUserCount can approximate the potential reach for a list of users.	integer

6.2.6 Upload files

POST /v1/messaging/upload/files

Description

Uploads a file to share it to user.

Type	Name	Description	Required
string	File	Base64 encoded data of image. Only image and video files are allowed. Jpg,png image, mpeg, mp4, h4v and webm video files are allowed.	Y

Response Status Codes

HTTP Code	Description	Schema
200	Success	
400	Bad Request: The server cannot or will not process the request due to an apparent client error (e.g., malformed request syntax, too large size, invalid request message framing, or deceptive request routing).	
401	Unauthorized: The request requires user authentication.	
500	Internal Server Error prevents the server from processing the request.	

Response Body

Name	Description	Schema
Fileid	Unique ID of the file uploaded	string

7. Callbacks/ Webhooks

This section defines the events and responses received from user to assistant. JBM will post events and responses to assistants at the webhook provided by the assistant.

Description

Webhook for user events, user messages, user actions and user capabilities callback.

Type	Name	Description	Schema
String	entityType	USER_EVENT, USER_MESSAGE, USER_ACTION, USER_CAPABILITIES	string
String	assistantId	ID of the assistant	string
String	userPhoneNumber	Phone number in E.164 format	string
String	Entity	Contains details of the entity described in entityType	UserMessage

Sample payload for all user messages and Clicks event payload:

a. User messages includes suggested action, suggested response and any p2a message

b. Sample USER_MESSAGE:

```
{
  "userPhoneNumber": "+9193218*****",
  "botId": "6544c5b408febf98e5fc5ec4",
  "entityType": "USER_MESSAGE",
  "entity": {
    "messageId":
      "MxTcYxYyXrTKS=o46jMjDJPQ",
    "sendTime": "2023-11-10T14:40:50.263298Z",
    "text": "",
    "userFile": null,
    "location": null,
    "suggestionResponse": {
      "postBack": {
        "data": "SR1L1C1"
      },
      "plainText": "",
      "type": "REPLY"
    }
  }
}
```

c. Sample P2A response:

```
{
  "userPhoneNumber": "+9193218*****",
  "botId": "6544c5b408febf98e5fc5ec4",
  "entityType": "USER_MESSAGE",
  "entity": {
    "messageId": "MxhVHQKZTxRkewsJQbKLsw9Q",
    "sendTime": "2023-11-10T14:38:38.105815Z",
    "text": "Hi",
    "userFile": null,
    "location": null,
    "suggestionResponse": null
  }
}
```

Apart from the above, JBM will be sending back REVOKE_SUCCESS and FAILURE which are called server events.

d. Sample server- event:

```
{
  "userPhoneNumber": "+9172004*****",
  "botId": "663a128464ee1034bb02ed7b",
  "entityType": "SERVER_EVENT",
  "entity": {
    "eventType": "TTL_EXPIRATION_REVOKED",
    "messageId": "77573603-2d32-481b-9442-c1ae5eace4a9",
    "sendTime": "2024-07-26T06:03:13.654Z",
    "phoneNumber": "+9172004*****",
    "eventId": "MxJ079=XIMR725MdGiofzEvA"
  }
}
```

e. Sample USER_EVENT:

```
{
  "userPhoneNumber": "+9198898*****",
  "botId": "651e8ec13800f6c7dee587c4",
  "entityType": "USER_EVENT",
  "entity": {
    "eventType": "MESSAGE_READ",
    "messageId": "ec ca1e-3293-49f9-896e-bbfaa3b13782",
    "sendTime": "2023-10-09T10:20:20.849997Z",
    "senderPhoneNumber": "+9198898*****",
    "eventId": "MxGP3uTVItTTmmTug5n9CBsw"
  }
}
```

There can be different USER_EVENTS. Namely, MESSAGE_READ, MESSAGE_DELIVERED, SUBSCRIBE and UNSUBSCRIBE events.

Unsubscribe indicates that the user has unsubscribed from receiving non-essential messages, such as promotions, from your Assistant. Users trigger this event by unsubscribing from the RBM conversation in Google Messages.

Subscribe indicates that a user wants to receive messages from your Assistant again, including non-essential content like promotions. Users can trigger this event by resubscribing to a conversation they previously unsubscribed from in Google Messages.

Example of SUBSCRIBE and UNSUBSCRIBE events:

```
{
  "userPhoneNumber":"+9132321*****",
  "botId":"6887547d0a11930ebc0debea",
  "entityType":"USER_EVENT",
  "entity": {
    "eventType":"UNSUBSCRIBE",
    "sendTime":"2025-07-02T09:43:29.749388Z",
    "senderPhoneNumber":"+9132321*****",
    "eventId":"02862972-2673-4664-b495-672e3a4203f1"
  }
}

{
  "userPhoneNumber":"+9132321*****",
  "botId":"6887547d0a11930ebc0debea",
  "entityType":"USER_EVENT",
  "entity": {
    "eventType":"SUBSCRIBE",
    "sendTime":"2025-07-02T09:43:29.749388Z",
    "senderPhoneNumber":"+9132321*****",
    "eventId":"02862972-2673-4664-b495-672e3a4203f1"
  }
}
```

The following table provides the file types allowed:

Media type	Document type	Extension	Works with rich cards
application/ogg	OGG audio	.ogg	No
application/pdf	PDF	.pdf	No
audio/aac	AAC audio	.aac	No
audio/mp3	MP3 audio	.mp3	No
audio/mpeg	MPEG audio	.mpeg	No
audio/mpg	MPG audio	.mp3	No
audio/mp4	MP4 audio	.mp4	No
audio/mp4-latm	MP4-latm audio	.mp4	No
audio/3gpp	3GPP audio	.3gp	No
image/jpeg	JPEG	.jpeg, .jpg	Yes
image/gif	GIF	.gif	Yes
image/png	PNG	.png	Yes
video/h263	H263 video	.h263	Yes
video/m4v	M4V video	.m4v	Yes
video/mp4	MP4 video	.mp4	Yes
video/mpeg4	MPEG-4 video	.mp4, .m4p	Yes
video/mpeg	MPEG video	.mpeg	Yes
video/webm	WEBM video	.webm	Yes

8. Schema Definitions

This section defines all the Data Transfer Objects (DTOs) being used in above sections.

CalendarEvent

Name	Description	Schema
<code>startTime</code>	Event start time. A timestamp in RFC3339 UTC "Zulu" format, with nanosecond resolution and up to nine fractional digits. Examples: "2023-06-26T15:01:23Z" and "2023-06-26T15:01:23.045123456Z"	string
<code>endTime</code>	Event end time. A timestamp in RFC3339 UTC "Zulu" format, with nanosecond resolution and up to nine fractional digits. Examples: "2023-06-26T15:01:23Z" and "2023-06-26T15:01:23.045123456Z".	string
<code>Title</code>	Event title.	string
<code>Description</code>	Event description.	string

Capabilities

Name	Description	Schema
features	enum Features	string

CardContent

Name	Description	Schema	Required
cardTitle	Title of the card	string	N
cardDescription	Description	string	N
cardMedia	Card media	Media	N
suggestions	Card suggestions. Max 4 suggestions	Array< Suggestion >	N

CarouselCards

Name	Description	Schema	Required
cardWidth	enum (SMALL_WIDTH, MEDIUM_WIDTH)	string	Y
contents	array of card contents	Array< CardContent >	Y

ContentInfo

Name	Description	Schema	Required
fileUrl	Publicly reachable URL of the file.	string	Y
thumbnailUrl	Publicly reachable URL of the thumbnail. Maximum size of 100 kB.	string	N

FileInfo

Name	Description	Schema
mimeType	MIME type of the file.	string
fileSizeBytes	Size of file (in bytes).	string
fileName	Name of the file.	string
fileUri	URI of the file.	string

Location

Name	Description	Schema
latitude	The latitude in degrees. It must be in the range [-90.0, +90.0].	number
longitude	The longitude in degrees. It must be in the range [-180.0, +180.0].	number

Media

Name	Description	Schema	Required
mediaHeight	enum (SHORT, MEDIUM, TALL)	string	Y
contentInfo	Content information	ContentInfo	Y

MessageContent

Name	Description	Schema	Required
Suggestions	Array of suggestions	Array< Suggestion >	Y
It's a Union field. It is only necessary to provide one of the following fields.			
Plaintext	Text of the message	string	N
contentInfo	Content information (file info)	ContentInfo	N
richCardDetails	Richcard details	RichCardDetails	N

Postback

Name	Description	Schema	Required
data	The base64-encoded string that the assistant receives in a user event when the user taps the suggested reply.	string	Y

RichCardDetails

Name	Description	Schema	Required
It's a Union field. It is only necessary to provide one of the following fields.			
Carousel		CarouselCards	
Standalone		StandaloneCard	

ShowLocation

Name	Description	Schema	Required
Location	Object representing the lat and longitude of the specified location	Location	Y
Label	(Optional) The label of the pin dropped at coordinates.	string	N

StandaloneCard

Name	Description	Schema	Required
cardOrientation	enum (HORIZONTAL, VERTICAL)	string	Y
content	Card content	CardContent	Y

SuggestedAction

Name	Description	Schema	Required
plainText	Text that is shown in the suggested action Max Length – 25 characters	string	Y
postback	Object containing postback information	Postback	Y
It's a Union field. It is only necessary to provide one of the following fields.			
dialerAction	Opens the user's default dialer app with the specified phone number filled in.	PhoneNumber	N
showLocation	Opens the user's default map app and selects the specified location	ShowLocation	N
openUrl	Opens the user's default web browser app to the specified URL. If the user has an app installed that is registered as the default handler for the URL, then this app will be opened instead, and its icon will be used in the suggested action UI.	Url	N
createCalendarEvent	Opens the user's default calendar app and starts the new calendar event flow with the Assistant-specified event data pre-filled	CalendarEvent	N
shareLocation	Opens the RCS app's location chooser so the user can pick a location to send back to the Assistant.	NA	N

SuggestedReply

Name	Description	Schema	Required
plaintext	Text that is shown in the suggested reply and sent back to the assistant when the user taps it. Max Length – 25 characters	string	Y
Postback	Object containing postback information	Postback	Y

Suggestion

Name	Description	Schema	Required
It's a Union field. It is only necessary to provide one of the following fields.			
Reply	Suggested reply	SuggestedReply	N
Action	Suggested action	SuggestedAction	N

SuggestionResponse

Name	Description	Schema
postbackData	The postback data specified by the Assistant when the Assistant sent the suggestion to the user.	string

text	The user-visible text of the suggestion that the user tapped.	string
type	The type of suggestion which was responded to.	enum (SuggestionType)

Url

Name	Description	Schema	Required
url	URL string	string	Y

UserFile

Name	Description	Schema
category	The category the file belongs to.	FileCategory
thumbnail	A thumbnail representing the file. Usually sent alongside image and video files. It's an Optional field.	FileInfo
payload	The media file	FileInfo

UserMessage

Name	Description	Schema
messageId	A unique message ID, assigned by the sending user's RCS client. Assistants should treat messages with the same (messageId, senderPhoneNumber) pair as unintended duplicates.	string
sendTime	A timestamp in RFC3339 UTC "Zulu" format, with nanosecond resolution and up to nine fractional digits. Examples: "2023-06-26T15:01:23Z" and "2023-06-26T15:01:23.045123456Z"	String
Text	Text, specifically a string created through organic user typing and not a suggested reply.	String
userFile	Media file	UserFile
location	Location. Note that this is not necessarily the user's location. A user is able to send arbitrary locations to an Assistant.	Location
suggestionResponse	Response generated by a user tapping a suggested reply or action.	suggestionResponse

8.1. ENUM Definitions

This section defines various ENUM variables referred in other sections.

Features

Name	Description
F_CHATASSISTANT_SUPPORTED	User supports chat assistant communication
F_REVOCATION_SUPPORTED	If the user supports revocation, the Assistant can revoke a message it sent before the JBM platform delivers the message
F_RICHCARD_SUPPORTED	Standalone rich cards
F_CAROUSEL_SUPPORTED	Carousel rich cards
F_ACTION_CREATE_CALENDAR	Suggested action to create calendar event
F_ACTION_DIAL_NUMBER	Suggested action to dial a number
F_ACTION_OPEN_URL	Suggested action to open a URL
F_ACTION_SHARE_LOCATION	Suggested action to share location
F_ACTION_VIEW_LOCATION	Suggested action to view location in a map app

FileCategory

Name	Description
FC_UNSPECIFIED	Not specified
FC_IMAGE	Image file (for example, png or gif)
FC_VIDEO	Video file (for example, mp4)
FC_AUDIO	Audio file (for example, mp3)

SuggestionType

Name	Description
ST_UNKNOWN	Unknown suggestion
ST_ACTION	Suggestion action
ST_REPLY	Suggested reply